

HOSPITAL DATA ANALYTICS

EXCEL DATA ANALYSIS & INTERACTIVE DASHBOARD

A DATA ANALYTICS PORTFOLIO PROJECT DEMONSTRATING
DATA CLEANING, ANALYSIS, VISUALIZATION, AND DASHBOARD
DEVELOPMENT USING MICROSOFT EXCEL.

WALTER ABAYA

DATA ANALYTICS • EXCEL • BUSINESS INTELLIGENCE

PROJECT OVERVIEW

TURNING HOSPITAL PATIENT DATA INTO ACTIONABLE INSIGHTS

PROJECT

Hospital Data Analytics

Interactive Excel dashboard analyzing patient demographics, admissions, length of stay, diagnoses, and hospital charges.

TOOL

Microsoft Excel

Excel Tables • Formulas • PivotTables • PivotCharts • Slicers • Timeline

FOCUS

Healthcare Analytics

Patient demographics • Hospital utilization • Admission trends • Financial analysis

OUTPUT

Interactive Dashboard

A dynamic dashboard allowing users to filter and explore hospital data by gender, age group, diagnosis, length of stay, and admission date.

PROJECT OBJECTIVE

TO TRANSFORM RAW HOSPITAL PATIENT DATA INTO A STRUCTURED, INTERACTIVE EXCEL DASHBOARD THAT SUPPORTS ANALYSIS OF PATIENT DEMOGRAPHICS, HOSPITAL UTILIZATION, ADMISSION TRENDS, DIAGNOSES, AND HOSPITAL CHARGES.

MY ROLE

DATA ANALYST — END-TO-END EXCEL PROJECT

- PREPARED AND VALIDATED THE DATASET
- CREATED CALCULATED FIELDS AND DATA-QUALITY CHECKS
- PERFORMED DESCRIPTIVE AND CATEGORICAL ANALYSIS
- BUILT PIVOTTABLES AND PIVOTCHARTS
- DESIGNED AN INTERACTIVE EXCEL DASHBOARD
- ADDED SLICERS AND TIMELINE FILTERING
- DOCUMENTED THE ANALYTICAL WORKFLOW

DATA PREPARATION & ANALYSIS

FROM RAW PATIENT RECORDS TO STRUCTURED ANALYTICAL DATA

SECTION 1 – DATA QUALITY

DATA QUALITY

- Duplicate Patient ID checks
- Date validation
- Patient age validation
- Length of stay validation
- Hospital charge validation
- Data-quality status checks

SECTION 2 – CALCULATED FIELDS

CALCULATED FIELDS

- Patient Age
- Age Group
- Length of Stay
- LOS Category
- LOS validation
- Charges validation

SECTION 3 – ANALYSIS

ANALYSIS

- Patient & admission counts
- Gender distribution
- Age-group distribution
- Top diagnoses
- Monthly admissions
- Monthly hospital charges
- Average length of stay
- Hospital charges by diagnosis



INTERACTIVE DASHBOARD

HOSPITAL PATIENT, UTILIZATION & FINANCIAL ANALYTICS



[README](#)

KEY FEATURES

7 KPI CARDS • 7 ANALYTICAL CHARTS • INTERACTIVE SLICERS • ADMISSION DATE TIMELINE

THE DASHBOARD PROVIDES AN INTERACTIVE VIEW OF PATIENT DEMOGRAPHICS, ADMISSIONS, HOSPITAL UTILIZATION, DIAGNOSES, AND FINANCIAL PERFORMANCE.

KEY INSIGHTS

WHAT THE ANALYSIS REVEALS

PATIENT PROFILE

The dashboard summarizes patient demographics across gender and age groups, providing a clear view of the hospital's patient population.

HOSPITAL UTILIZATION

Length-of-stay analysis highlights how patients are distributed across same-day, short-stay, and longer-stay categories.

ADMISSION TRENDS

Monthly analysis allows hospital activity to be monitored over time, helping identify changes in admission volume and hospital charges.

FINANCIAL ANALYSIS

Hospital charges were analyzed by month, diagnosis, and patient characteristics to provide a financial perspective on hospital activity.

ANALYTICAL VALUE

THE DASHBOARD TRANSFORMS DETAILED PATIENT-LEVEL RECORDS INTO A CONCISE MANAGEMENT VIEW, MAKING IT EASIER TO IDENTIFY PATTERNS IN PATIENT VOLUME, UTILIZATION, DIAGNOSES, AND HOSPITAL CHARGES.

SKILLS & PROJECT OUTCOME

EXCEL CAPABILITIES DEMONSTRATED THROUGH THE PROJECT

DATA PREPARATION

- DATA CLEANING
- DATA VALIDATION
- DUPLICATE CHECKS
- DATA QUALITY CONTROLS

EXCEL ANALYSIS

- COUNTIF / COUNTA
- SUM / AVERAGE
- IF / AND / OR
- DATEDIF
- CALCULATED FIELDS

DATA VISUALIZATION

- PIVOTTABLES
- PIVOTCHARTS
- KPI CARDS
- INTERACTIVE DASHBOARD

INTERACTIVITY

- SLICERS
- TIMELINE FILTERING
- DASHBOARD NAVIGATION

PROJECT OUTCOME

DEVELOPED A COMPLETE EXCEL-BASED HOSPITAL ANALYTICS SOLUTION THAT TRANSFORMS PATIENT-LEVEL DATA INTO AN INTERACTIVE DASHBOARD FOR EXPLORING DEMOGRAPHICS, ADMISSIONS, HOSPITAL UTILIZATION, DIAGNOSES, AND FINANCIAL ACTIVITY.

TURNING DATA INTO INSIGHTS THROUGH EXCEL

WALTER ABAYA

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